



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	VeriCode AI: Adaptive AI-Powered DevSecOps and Software Quality Management Platform	
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Abstract

VeriCode AI is an AI-powered Adaptive DevSecOps and Software Quality Management Platform designed to assist development teams throughout the Software Development Life Cycle. The platform integrates Artificial Intelligence, Agile methodologies, DevOps automation, security analysis, and MLOps practices into a unified environment. It uses an LLM-based AI component to analyze source code, identify coding issues and potential vulnerabilities, explain detected problems, and provide intelligent recommendations for improving code quality and security. Agile practices such as user stories, backlog management, sprint planning, and task tracking are incorporated to support structured software development.

The platform integrates DevOps and DevSecOps practices to automate and secure the software development pipeline. Git and GitHub are used for version control, while GitHub Actions enables CI/CD automation for building, testing, security analysis, and deployment. SonarQube is used for static code analysis, Trivy for vulnerability scanning, and OWASP ZAP for dynamic application security testing. Docker and Kubernetes support containerized application deployment and orchestration, while Prometheus and Grafana provide monitoring and visualization of application and pipeline metrics. These integrations enable security and quality issues to be identified earlier in the development lifecycle and reduce the effort involved in manual analysis and deployment.

VeriCode AI further incorporates MLOps concepts for managing, deploying, and monitoring its AI components and uses feedback from code quality, security findings, project progress, and system monitoring to provide adaptive and context-aware recommendations. Based on the detected project conditions, the platform can prioritize issues, recommend corrective actions, and support development and deployment decisions. By combining Agile, AI, DevOps, DevSecOps, MLOps, containerization, security testing, and observability, the proposed system aims to improve software quality, security, development efficiency, and reliability while demonstrating the practical application of modern software engineering practices.

Signature of the Guide

HOD